Lakshya JEE 2.0 2025

Chemistry

P Block Elements

DPP: 7

Q1 Which of the following hydrides has the lowest boiling point?
(A) H_2O
(B) H_2S
(C) H_2Se
(D) H_2Te

Q2 Which of the following shows the highest boiling point
(A) H_2O
(B) H_2S
(C) H_2Se
(D) H_2Te

Q3 $S-S$ bond is present in
(A) $H_2S_2O_7$
(B) $H_2S_2O_8$
(C) H_2S_2O6
(D) H_2SO_5

Q4 Roasting of sulphides give the gas X as a byproduct. This is a colourless gas with cooking smell of burnt sulphur and causes great damage to respiratory organs as a result of acid rain. Its aqueous solution is acidic, acts as a reducing agent and its acid has never been isolated. The gas X is
(A) SO_2
(B) SO_3
(C) CO_2
(D) H_2S

Q5 Oleum is a mixture of
(A) $H_2SO_4+SO_2$
(B) $H_2SO_4+SO_3$
(C) $H_2\ S_2O_3+SO_3$
(D) $H_2S_2O_3+SO_2$

Q6 The acidic character of dioxides of members of oxygen family decreases in the order
(A) $SeO_2>SO_2>TeO_2>PoO_2$
(B) $SO_2>SeO_2>TeO_2>PoO_2$
(C) $PoO_2>TeO_2>SeO_2>SO_2$
(D) $TeO_2>PoO_2>SeO_2>SO_2$

Q7 What is the basicity of peroxodisulfuric acid?
(A) 1
(B) 3
(C) 4
(D) None of these

Q8 The colour shown by halogen is incorrectly given by
(A) F_2 - Yellow
(B) Cl_2 - Colourless
(C) Br_2 - Red
(D) I_2 - Violet

Q9 The halogen, which is solid at room temperature is
(A) I_2
(B) Br_2
(C) Cl_2
(D) F_2

Q10 Deacon's process of manufacture of chlorine is represented by the equation
(A) $MnO_2+HCl\rightarrow MnCl_2+Cl_2+H_2O$
(B) $KMnO_4+HCl\rightarrow KCl+MnCl_2+H_2O+Cl_2$
(C) $HCl+O_2\xrightarrow{CuCl_2}Cl_2+H_2O$
(D) $NaCl+MnO_2+H_2SO_4\rightarrow Cl_2+MnCl_2+NaHSO_4+H_2O$

Q11 Which of the following represents outermost shell electronic configuration of halogens
(A) s^2p^3
(B) s^2p^6

(C) s^2p^4
(D) s^2p^5

Q12 Which one of the following halogens liberates oxygen on reacting with H_2O
(A) I_2
(B) Cl_2
(C) Br_2
(D) F_2

Q13 In which case, the order of acidic strength is not correct?
(A) $HI > HBr > HCl$
(B) $HIO_4 > HBrO_4 > HClO_4$
(C) $HClO_4 > HClO_3 > HClO_2$
(D) $HF > H_2O > NH_3$

Q14 The shape of ClO_3^- is
(A) Pyramidal
(B) Tetrahedral
(C) Triangular planar
(D) Triangular bipyramidal

Q15 Which is incorrectly given according to order indicated?
(A) $F_2 > Cl_2 > Br_2 > I_2$; Oxidising power
(B) $HI > HBr > HCl > HF$; Acidic strength
(C) $F_2 > Cl_2 > Br_2 > I_2$; Bond dissociation energy
(D) $HF > HI > HBr > HCl$; Boiling point

Q16 Which of the following displaces Br_2 from an aqueous solution containing bromide ions?
(A) I_2
(B) I_3^-
(C) Cl_2
(D) Cl^-

Q17 The shape of I_3^- is
(A) Tetrahedral
(B) Linear
(C) T-shape
(D) Trigonal

Q18 Which anion can undergo both oxidation and reduction?
(A) $Cr_2O_7^{2-}$
(B) NO_3^-
(C) OCl^-
(D) S^{2-}

Answer Key

Q1	(B)	Q10	(C)
Q2	(A)	Q11	(D)
Q3	(C)	Q12	(D)
Q4	(A)	Q13	(B)
Q5	(B)	Q14	(A)
Q6	(B)	Q15	(C)
Q7	(D)	Q16	(C)
Q8	(B)	Q17	(B)
Q9	(A)	Q18	(C)

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